Test 10:

READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

Deep sea discovery

Recent research has provided new insights into how fish communicate.

Nico Michiels is an ecologist from the University of Tübingen in Germany who spends part of each year in Egypt, where he dives in the Red Sea, observing fish life and gathering data on its coral reefs. In September 2007 he decided to find out how far red light could penetrate the ocean depths. Seawater absorbs different colours at different depths, and as an experienced diver, Michiels was aware that red light is extinguished not far below the surface whereas blue-green light penetrates deeper. To find out the depth at which red disappeared in this particular ocean, however, he attached a special plastic filter to his dive mask, which was designed to block out all colours except red. Then he began to descend.

In theory, once he reached about 15 metres, he should have been plunged into darkness. Instead, something totally unexpected happened. Sure enough, 20 metres down it was as dark as night. ‘All the fish disappeared. With no light from the surface, they were effectively black and had become invisible,’ he says. ‘But it didn’t stay black for long. Then I saw a group of goby fish with bright red eyes lit up against the background. After that, red spots began to show up all over the reef.’

Even with the red filter removed, Michiels could pick them out without much trouble once his eyes grew accustomed to the gloom. It seems strange that no diver or researcher had spotted all this red before, but as Michiels points out, no one saw it because no one expected to see it. On that one dive, Michiels discovered three fish species with prominent red markings, and has found many others since.

But how can fish appear red where there's no red light? Ordinary red pigments* look red because they reflect red light while absorbing all other wavelengths. At 20 metres down, there had to be some other explanation for the red Michiels was seeing. He suspected fluorescence. Fluorescent pigments behave differently from ordinary ones: they receive incoming light of one wavelength, for example blue, and emit light of a longer wavelength, in this case red. On the reef in the Red Sea during daytime, the most likely explanation was that the predominantly blue and green wavelengths at depth triggered the emission of fluorescent red in the fish.

With only a week left in Egypt, and lacking the equipment to confirm that the fish were fluorescent, Michiels photographed as many of them as he could. Then, once back in Germany, he bought an assortment of tropical fish and installed them in his lab. Here he confirmed that the fish did indeed fluoresce. In most of the fish he looked at, the fluorescence could be traced to specialised pigment cells that lie in the skin beneath the scales. These cells contain 'guanine crystals', which scatter light to give fish their silvery sheen. However, Michiels says they are still not sure exactly what is fluorescing. 'It's not the crystals themselves. It's probably a fluorescent protein built into the crystals, and we have a suspicion that it might be made by bacteria.'

Intrigued, Michiels began a systematic search for red fluorescence in reef fish. He and his colleagues, Nils Anthes and Dennis Sprenger, have identified some 50 species with red fluorescence. The most common markings tend to be on the body towards the head and to a lesser extent around the eyes, and then the fins. To Michiels, the distribution of these markings is one of the strongest indications that red fluorescence has a very particular function: communication with other members of the species.

According to several recent studies, a whole range of animals employ fluorescence as a natural highlighter to boost the visibility of body parts they use to signal, for example to ward off enemies. In reef fish, the red tends to be confined to parts of the body used to signal, suggesting these markings serve a similar function. But instead of highlighting an existing colour, the fluorescence gives the fish a colour that otherwise wouldn't exist.

For example, fish commonly use eye rings to signal that they are present and their direction of gaze, and Michiels suspects that red-eyed gobies use signals to indicate their location and keep their group together.

Red light, whatever its source, doesn't travel far through water, which suggests signals are intended to be private, seen only by nearby fish of the right species. There are several lines of evidence to support this, says Michiels. And closely related species do not have completely identical markings, which suggests they might be important in species recognition.

Michiels suspects red fluorescence has another important role for some reef fish: helping them blend in. During his first dive with the red filter, he noticed corals glow a dark but faint red too. Against this irregular red background, a fish that glows red all over would be hard to distinguish. More compelling for Michiels is the case of the scorpionfish, which lies perfectly still until food swims past, which it then sucks in.

Yet if red plays any part in a fish's life, then it must be able to see it. Fish that live in a world dominated by blue-green light are assumed to have eyes tuned to those wavelengths, and most marine fish that have been studied are thought incapable of seeing red. One exception is the seahorse, whose eyes are sensitive to red. As for the other fish, it remains to be seen.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–6 on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

1. During his 2007 dive, Michiels expected to encounter total darkness at about 15 metres.

2. Michiels could see the red markings on fish without the aid of the red filter.

3. Other divers had assumed they would see fish with red markings.

4. All the fish with red markings that Michiels found during his diving expeditions came from the Red Sea.

5. Michiels first thought of the possibility that fish could fluoresce while he was in Germany.

6. Michiels remains uncertain as to what creates fluorescence in fish.

Questions 7–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 7–13 on your answer sheet.

Michiels's findings

Michiels has observed:

- 50 types of fish with red fluorescence in total
- markings mainly near the **7.** _____

Some of Michiels's beliefs are that:

- red fluorescence is used specifically for **8.** _____ purposes
- fish, like some animals, use fluorescence to keep **9.** _____ away
- gobies depend on red fluorescence to show their **10.** _____
- there are variations in the markings of fish among those **11.** _____ which are very similar

Other benefits of red fluorescence:

- fish cannot easily be seen near backgrounds of **12.** _____ which give off a red light
- helps some fish catch their prey

The ability to see red amongst fish:

- the only fish proven to have this ability is the **13.** _____

ĐÁP ÁN:

1. TRUE (Đoạn 1: Theo lý thuyết, một khi ông chạm đến độ sâu khoảng 15 mét, ông lẽ ra đã bị chìm vào bóng tối).

Đoạn chứa đáp án:

In theory, once he reached about 15 metres, he should have been plunged into darkness.

2. TRUE (Đoạn 2: Ngay cả khi đã tháo kính lọc màu đỏ, Michiels vẫn có thể nhận ra chúng mà không gặp nhiều rắc rối khi mắt đã quen với bóng tối).

Đoạn chứa đáp án:

After that red spots began to show up all over the reef.'

3. FALSE (Đoạn 2: Michiels chỉ ra rằng không ai nhìn thấy sắc đỏ này trước đây vì không ai **mong đợi** sẽ thấy nó).

Đoạn chứa đáp án:

It seems strange that no diver or researcher had spotted all this red before, but as Michiels points out, no one saw it because no one expected to see it.

4. NOT GIVEN (Bài đọc nói ông tìm thấy 3 loài ở Biển Đỏ trong lần lặn đầu tiên, nhưng không khẳng định **tất cả** các loài ông tìm thấy sau này đều ở đó).

Đoạn chứa đáp án:

On that one dive Michiels discovered three fish species with prominent red markings, and has found many others since.

5. FALSE (Đoạn 3: Khi còn ở rạn san hô tại Biển Đỏ, ông đã nghi ngờ hiện tượng huỳnh quang là lời giải thích hợp lý nhất).

6. TRUE (Đoạn 1: Michiels nói rằng họ vẫn chưa chắc chắn chính xác cái gì đang phát huỳnh quang).

Đoạn chứa đáp án:

However, Michiels says they are still not sure exactly what is fluorescing.

7. head (Các dấu vết phổ biến nhất có xu hướng nằm trên cơ thể về phía đầu).

Đoạn chứa đáp án:

The most common markings tend to be on the body towards the head and to a lesser extent around the eyes, and then the fins.

8. communication (Sự phân bố các dấu vết này là chỉ dấu mạnh mẽ cho chức năng cụ thể: giao tiếp với các thành viên cùng loài).

Đoạn chứa đáp án:

To Michiels, the distribution of these markings is one of the strongest indications that red fluorescence has a very particular function: communication with other members of the species.

9. enemies (Sử dụng huỳnh quang như một công cụ làm nổi bật tự nhiên để cảnh báo kẻ thù tránh xa).

Đoạn chứa đáp án:

According to several recent studies, a whole range of animals employ fluorescence as a natural highlighter to boost the visibility of body parts they use to signal, for example to ward off enemies.

10. location (Ông nghi ngờ cá bống mắt đỏ sử dụng tín hiệu để chỉ ra vị trí của chúng và giữ cho nhóm ở cùng nhau).

Đoạn chứa đáp án:

and Michiels suspects that red-eyed gobies use signals to indicate their location.

11. species (Các loài có quan hệ họ hàng gần không có các dấu vết hoàn toàn giống nhau, giúp nhận dạng loài).

Đoạn chứa đáp án:

And closely related species do not have completely identical markings , which suggests they might be important in species recognition.

12. corals (Trong lần lặn đầu tiên, ông nhận thấy san hô cũng phát ra ánh sáng đỏ tối và mờ nhạt, giúp cá ẩn nấp).

Đoạn chứa đáp án:

During his first dive with the red filter, he noticed corals glow a dark but faint red too. Against this irregular red background, a fish that glows red all over would be hard to distinguish.

13. seahorse (Ngoại lệ duy nhất là cá ngựa, loài có mắt nhạy cảm với ánh sáng đỏ).

Đoạn chứa đáp án:

most marine fish that have been studied are thought incapable of seeing red. One exception is the seahorse, whose eyes are sensitive to red.

TRUE IELTS

READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

The conquest of malaria in Italy

A review of Frank M Snowden's masterly history of a country's fight to eradicate a deadly disease

A The word 'malaria' means 'bad air' in Italian, and this terrible disease marked the life of the people of that country for thousands of years. Yet by 1962, Italy was officially declared malaria-free, and it has remained so ever since. Frank Snowden's study of this successful endeavour is a remarkable piece of historical work. Original, crystal clear, analytical and passionate, Snowden takes us to areas historians have rarely visited before.

B Everybody now knows that malaria is carried by mosquitoes. But in the 19th century most experts subscribed to the theory of 'miasma' or 'poisoning of the air'. Others made a link between swamps, water and malaria, but did not make the further leap towards insects. The consequence of these theories was that little was done to combat the disease before the end of that century. The situation was so serious that from a total population of 25m Italians, 11m were 'permanently at risk'. In warm, damp, malarial zones, the life expectancy of land workers was a terrifying 22.5 years. The economic impact of the disease was immense. Epidemics were blamed on people who originated from the hotter parts of Italy, given the widespread belief that malaria was hereditary.

C One of the first breakthroughs in the war against malaria came in 1898 when the zoologist Giovanni Battista Grassi demonstrated that the micro-organisms causing the disease were carried in the digestive tract of the mosquito. By releasing mosquitoes into rooms to drink the blood of healthy human volunteers, Grassi was able to make the direct link between the insects and the disease. Definitive proof of this theory was obtained after an extraordinary series of experiments in Italy, where healthy people were introduced into malarial zones but kept free of mosquito bites – and remained well. The recently formed Italian state at last had the necessary information to begin tackling the disease.

D A complicated approach was adopted, which made use of quinine – a drug obtained from tree bark which had long been used to combat fever, but was now seen as a crucial part of the war on malaria. Italy introduced a quinine law and a quinine tax in 1904, and the drug was administered to large numbers of rural workers. Despite its often terrible side effects, the drug was successful in limiting the spread of the disease, and in breaking cycles of infection. In addition, Italy set up rural health centres and invested heavily in education programmes. Malaria, as Snowden shows, was not just a medical problem, but a social and regional issue, and could only be defeated through multi-layered strategies.

E It was originally decided to give quinine to all those in vulnerable regions – even healthy people. However, peasants were often suspicious of medicine being forced upon them, and doctors were frequently met with hostility and stubborn refusal to accept the treatment offered. But despite the refusals, the strategy as a whole was hugely successful. Deaths

from malaria fell by some 80% in the first decade of the 20th century and some areas escaped altogether from the scourge of the disease.

F The 1915-18 war delayed the campaign, as funds were diverted to the battlefields. In the 1920s and 1930s the draining of the damp, unhealthy marshlands around Rome had a certain impact on the spread of malaria. However, as war swept through the drained lands in the 1940s, the disease returned with a vengeance. Yet the country's leading malariologist Alberto Missiroli refused to order the distribution of quinine, so allowing the epidemic to spread unchecked. According to Snowden, he did this in order to create the ideal conditions for a new strategy, supported by the US Rockefeller Foundation – a massive experiment involving the extermination of mosquitoes with the chemical DDT. It is estimated that more than a third of the inhabitants in the affected area contracted malaria and countless thousands died.

G With the end of the war in 1945, the US government and the Rockefeller Foundation were free to experiment. DDT was sprayed from the air, and 3m Italians had their bodies covered with the chemical. The effects were immense, and by 1962 malaria was more or less gone from the whole country. One of the final victims to die of the disease in Italy was the popular cyclist, Fausto Coppi. He had contracted malaria in Africa in 1960, and the failure of doctors in Italy to spot the disease was a sign of the times. A few decades earlier, they would have immediately noticed the telltale signs; it was later claimed that a small dose of quinine would have saved his life.

H As there are still more than 1m deaths every year from malaria worldwide, Snowden's book also has contemporary relevance. This is a disease that affects every level of the societies where it is rampant. The economic miracle of the 1950s and 1960s which made Italy into a modern industrial nation would not have been possible without the eradication of malaria. Moreover, this book convincingly argues that the disease was 'an integral part of the big picture of modern Italian history'. This magnificent study, beautifully written and impeccably documented, deserves an audience beyond specialists in history, or in Italy.

Questions 14–17

Complete the summary below.

*Choose **NO MORE THAN TWO WORDS** from the passage for each answer.*

Write your answers in boxes 14–17 on your answer sheet.

Malaria in 19th-century Italy

Up until the late nineteenth century, experts failed to make the connection between malaria and **14** ..... . The most popular belief at the time was the **15** ..... theory, which upheld the idea that diseases were carried by unclean air. Another common idea was that malaria was a **16** ..... disease, and as a result people from certain parts of the country were often held responsible for the spread of epidemics. Malaria was particularly widespread in rural regions, where **17** ..... could be extremely short.

Questions 18–20

Do the following statements agree with the information given in Reading Passage 2?

In boxes 18–20 on your answer sheet, write

TRUE *if the statement agrees with the information*
FALSE *if the statement contradicts the information*
NOT GIVEN *if there is no information on this*

18 The volunteers in Giovanni Battista Grassi's research came from all parts of Italy.

19 Experiments in Italy proved that it was possible to remain healthy despite being in malarial zones.

20 In the early twentieth century, quinine was successfully administered to all inhabitants of vulnerable regions.

Questions 21–26

Reading Passage 2 has eight paragraphs, ***A–H****.*

Which paragraph contains the following information?

Write the correct letter, ***A–H****, in boxes 21–26 on your answer sheet.*

21 a figure showing the dramatic results of an anti-malarial drug programme

22 an important discovery about how malaria is spread

23 mention of an expert's decision not to halt the spread of the disease

24 the significance of the malaria story for today's readers

25 examples of false assumptions which held back scientific understanding of malaria

26 a reference to legislation to support the fight against malaria

TRUE IELTS

ĐÁP ÁN:

14. insects: Cho đến cuối thế kỷ 19, các chuyên gia vẫn chưa tìm ra mối liên hệ giữa sốt rét và **côn trùng** (cụ thể là muỗi).

Đoạn chứa đáp án:

Others made a link between swamps, water and malaria, but did not make the further leap towards insects.

15. miasma: Niềm tin phổ biến nhất thời bấy giờ là thuyết **"miasma" (chướng khí)**, cho rằng bệnh lây truyền qua không khí ô nhiễm.

Đoạn chứa đáp án:

But in the 19th century most experts subscribed to the theory of 'miasma' or 'poisoning of the air'.

16. hereditary: Một ý tưởng phổ biến khác là sốt rét là bệnh **di truyền**.

Đoạn chứa đáp án:

Epidemics were blamed on people who originated from the hotter parts of Italy, given the widespread belief that malaria was hereditary.

17. life expectancy: Tại các vùng nông thôn có sốt rét, **tuổi thọ** trung bình của người lao động cực kỳ ngắn (chỉ khoảng 22.5 tuổi).

Đoạn chứa đáp án:

In warm, damp, malarial zones, the life expectancy of land workers was a terrifying 22.5 years.

18. NOT GIVEN: Bài viết chỉ đề cập đến "những tình nguyện viên khỏe mạnh", không nói rõ họ có đến từ khắp mọi miền nước Ý hay không.

Đoạn chứa đáp án:

By releasing mosquitoes into rooms to drink the blood of healthy human volunteers, Grassi was able to make the direct link between the insects and the disease.

19. TRUE: Các thí nghiệm đã chứng minh rằng người ta vẫn có thể khỏe mạnh trong vùng sốt rét nếu tránh được muỗi đốt.

Đoạn chứa đáp án:

Definitive proof of this theory was obtained after an extraordinary series of experiments in Italy, where healthy people were introduced into malarial zones but kept free of mosquito bites - and remained well.

20. FALSE: Quinine không được dùng cho tất cả cư dân vì nhiều nông dân nghi ngờ và từ chối điều trị.

Đoạn chứa đáp án:

Italy introduced a quinine law and a quinine tax in 1904, and the drug was administered to large numbers of rural workers.

21. E: Một số liệu cho thấy kết quả ấn tượng của chương trình thuốc chống sốt rét (tỷ lệ tử vong giảm 80% trong thập kỷ đầu thế kỷ 20).

Đoạn chứa đáp án:

But despite the refusals, the strategy as a whole was hugely successful. Deaths from malaria fell by some 80% in the first decade of the 20th century and some areas escaped altogether from the scourge of the disease.

22. C: Một khám phá quan trọng về cách sốt rét lây lan (Giovanni Battista Grassi chứng minh vi sinh vật gây bệnh nằm trong đường tiêu hóa của muỗi).

Đoạn chứa đáp án:

One of the first breakthroughs in the war against malaria came in 1898 when the zoologist Giovanni Battista Grassi demonstrated that the micro-organisms causing the disease were carried in the digestive tract of the mosquito.

23. F: Đề cập đến quyết định của một chuyên gia không ngăn chặn sự lây lan của bệnh (Alberto Missiroli từ chối phân phối quinine để tạo điều kiện cho chiến lược mới với DDT).

Đoạn chứa đáp án:

Yet the country's leading malariologist Alberto Missiroli refused to order the distribution of quinine, so allowing the epidemic to spread unchecked. According to Snowden, he did this in order to create the ideal conditions for a new strategy, supported by the US Rockefeller Foundation – a massive experiment involving the extermination of mosquitoes with the chemical DDT.

24. H: Ý nghĩa của câu chuyện sốt rét đối với độc giả ngày nay (tầm quan trọng đương đại khi thế giới vẫn có hơn 1 triệu ca tử vong mỗi năm).

Đoạn chứa đáp án:

As there are still more than 1m deaths every year from malaria worldwide, Snowden's book also has contemporary relevance.

25. B: Các ví dụ về những giả định sai lầm đã cản trở hiểu biết khoa học về sốt rét (thuyết chướng khí và quan niệm bệnh di truyền).

Đoạn chứa đáp án:

But in the 19th century most experts subscribed to the theory of 'miasma' or 'poisoning of the air'. Others made a link between swamps, water and malaria, but did not make the further leap towards insects. The consequence of these theories was that little was done to combat the disease before the end of that century.

26. D: Đề cập đến luật pháp để hỗ trợ cuộc chiến chống sốt rét (nước Ý ban hành luật và thuế quinine năm 1904).

Đoạn chứa đáp án:

In addition, Italy setup rural health centres and invested heavily in education programmes. Malaria, as Snowden shows, was not just a medical problem, but a social and regional issue, and could only be defeated through multi-layered strategies.

READING PASSAGE 3

*You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 below.*

Science and Filmmaking

Academics are now working more with filmmakers who are impressed by the results of their research in computer-generated imagery (CGI)

Every year the film academy in the USA celebrates the outstanding achievements of the year in a ceremony known as the Oscars. An increasingly important component of the ceremony is the presentation of the Scientific and Technical awards. In 2004 a notable event took place: the academic world met the cinematographic world when researchers from Stanford University in the USA were awarded an Oscar. These researchers, led by Steve Marschner, were from the field of Computer Graphics at Stanford. They were part of a growing cohort of computer scientists that has become fundamental to moviemaking.

Films have shown that it is possible to use CGI to make actors look younger, older, weaker or stronger than they actually are in a surprisingly realistic manner. At least, it is possible if the altered actors are not filmed too closely. This is because the difficulty of recreating the textures of both skin and fabric means that the effect is less convincing when seen close up. The work of Marschner and his colleagues has greatly improved the accurate and realistic modeling of both skin and fabric. They recognized that one of the difficulties of creating lifelike characters in the computer world is that, in CGI, the characters' skin is opaque (two-dimensional) whereas real skin is in fact translucent (three-dimensional), that is to say, it is semi-transparent.

Marschner and his colleagues received the Oscar for their work in successfully producing a CGI model that simulates translucency; this is when light penetrates skin and then scatters below the skin's surface before re-emerging. This is called subsurface scattering, and the mathematics for the model goes back many decades to the time when it was used in astrophysics. Because human skin is naturally translucent, it was necessary to be able to create this artificially in order to simulate the soft appearance of real skin. Previous CGI models, which assumed that skin was entirely opaque, resulted in characters with a plastic appearance. The scientists' new model of CGI was so important in bringing digital characters to life that, within two years of their original research paper, all the major special-effects studios had incorporated it into their digital rendering systems.

However, despite their award, the scientists, with admirable tenacity, continued their search for perfection, as they still did not feel that they had yet satisfactorily recreated the subtle ways light is reflected. To do this, they began to look in detail at the way skins and fabrics reflect light differently according to their make-up: the exact arrangement of fibers in fabric and the network of fibers in skin. Marschner and his team tried to do this by using computerized tomography, which is most familiar as a medical technique for examining people's internal organs. Like classical radiography, it uses X-rays, but because the image is constructed inside a computer using exposures taken from many different positions, rather

than a single exposure on photographic film, it can capture fine details that are invisible in classical radiography.

Unfortunately, the scientists understood that at this point in time they could not use computerized tomography on skin, because a very high-intensity X-ray is needed to show the kind of detail they wanted and this would be very dangerous for human skin. They have, however, had some success with fabric. Using this new method of imaging, they are able to accurately record the three-dimensional structure of all the fibers in a number of small pieces of fabric. These same pieces of fabric, through the use of CGI, can then be patched together to form an entire garment inside a computer, in the same way that a small group of actors can be made to look like hundreds of people gathered together. A garment created through CGI is therefore made up of pieces of fabric whose internal structure has been pre-recorded. This means that the way light is reflected by the garment can be calculated far more realistically than if the scientists just made a computer model of what they thought the interior of the fabric looked like. Cinematography will benefit from this because, although it may take some years to be able to use computerized tomographic imaging of skin, for the moment the movement of a virtual cloak or the lifting of a computerized hat should look far more realistic.

In the meantime, according to Marschner's colleague Kavita Bala, the technology might have an application in online retailing. At the moment, people buying clothes over the internet have only a standard photograph to help them choose their purchases. It is hoped that if online shoppers can view items which have been presented through the use of computerized tomography graphics, they will have a much better understanding of what the material the item is made of is really like.

Marschner is now working on the way light is scattered from individual hairs. He says, 'I feel lucky to be working in this niche. I'm a visual person and to be able to spend my time scrutinizing the world around me, trying to understand why it looks the way it does, is very rewarding'.

Questions 27–32

Choose the correct letter, ***A, B, C, or D****.*

Write the correct letter in boxes 27–32 on your answer sheet.

27 What is the writer's main point in the first paragraph?

A Computer scientists are rarely represented at the Oscars.
B Film-industry awards hold little interest for computer scientists.
C The USA rewards film actors more than computer scientists.
D Computer scientists are becoming a vital part of the film industry.

28 When describing the way computer-generated imagery changes actors' appearance, what does the writer suggest?

A CGI looks best from a distance.
B CGI interferes with actors' facial expressions.
C Audiences expect too much of CGI.
D The scientists had hoped for more convincing results.

29 What does the writer suggest about the scientists' attitude to their work in the fourth paragraph?

A They were motivated to get as near to reality as possible.
B They were interested in gaining recognition for their work.
C They thought it could have medical applications.
D They hoped to receive further funding for their research.

30 What are we told about computerised tomography?

A It has only recently been used by doctors.
B It is similar to X-rays in the way it works.
C Filmmakers have used it for many years for special lighting effects.
D It can be administered using traditional radiography machines.

31 Which of these advantages does the writer attribute to the current use of computerised tomography?

A It is most effective when used to create images of skin.
B It can make a few people seem like a crowd.
C It allows clothes designers to create new designs.
D It is practical because of the time it takes.

32 The writer mentions Kavita Bala in order to

A comment on the success of CGI in commercial contexts.
B highlight the link between CGI and photography.

C show that Marschner's team are uncertain about the future of CGI.
D demonstrate a use for CGI outside the film industry.

Questions 33–36

Do the following statements agree with the views of the writer in Reading Passage 3?

In boxes 33–36 on your answer sheet, write

YES — if the statement agrees with the views of the writer
NO — if the statement contradicts the views of the writer
NOT GIVEN — if it is impossible to say what the writer thinks about this

33 It used to be unusual for university researchers to receive a cinematography award.

34 CGI is popular among ageing actors.

35 The scientists' success in generating images of complete CGI garments has won them many awards.

36 It will be a long time before computerised tomographic *imaging of fabric* is used by filmmakers.

TRUE IELTS

Questions 37–40

*Complete the summary using the list of words, **A–I**, below.*

*Write the correct letter, **A–I**, in boxes 37–40 on your answer sheet.*

The work of Marschner and his colleagues

For many years, CGI characters did not appear entirely lifelike as their skin appeared plastic. Marschner and his colleagues were the first to apply an understanding of how **37** _____ interacts with human skin. Their CGI model is based on a novel application of principles of **38** _____, which had previously been applied in other scientific research. The importance of CGI to the film industry has led to the **39** _____ of Marschner's model by special-effects studios. Marschner's model has led to the **40** _____ of cinematography.

A light
B transparency
C age
D use
E astrophysics
F mathematics
G improvement
H colour
I translucency

TRUE IELTS

ĐÁP ÁN:

27. D: Điểm chính của tác giả ở đoạn đầu là các nhà khoa học máy tính đang trở thành một phần thiết yếu của ngành công nghiệp điện ảnh (Đoạn 1: "...computer scientists that has become fundamental to moviemaking").

28. A: Khi mô tả cách CGI thay đổi ngoại hình diễn viên, tác giả gợi ý rằng CGI trông đẹp nhất khi nhìn từ xa (Đoạn 2: "...it is possible if the altered actors are not filmed too closely...").

29. A: Ở đoạn 4, thái độ của các nhà khoa học đối với công việc là họ có động lực để đạt được kết quả gần với thực tế nhất có thể (Đoạn 4: "...continued their search for perfection...").

30. B: Về kỹ thuật chụp cắt lớp vi tính (CT), bài viết cho biết nó tương tự như tia X về cách thức hoạt động (Đoạn 4: "Like classical radiology, it uses X-rays...").

31. B: Ưu điểm của việc sử dụng CT hiện nay là nó có thể làm cho một vài người trông giống như một đám đông (Đoạn 5: "...in the same way that a small group of actors can be made to look like hundreds of people...").

32. D: Tác giả nhắc đến Kavita Bala để chứng minh một ứng dụng của CGI bên ngoài ngành công nghiệp điện ảnh (Đoạn 6: nhắc đến ứng dụng trong bán lẻ trực tuyến - online retailing).

33. YES: Trước đây, việc các nhà nghiên cứu đại học nhận được giải thưởng điện ảnh là điều không bình thường (Đoạn 1: mô tả sự kiện năm 2004 là "notable" - đáng chú ý/khác biệt).

34. NOT GIVEN: Không có thông tin về việc CGI có phổ biến đối với các diễn viên lớn tuổi hay không.

35. NOT GIVEN: Không có thông tin về việc các nhà khoa học giành được "nhiều giải thưởng" nhờ tạo ra hình ảnh trang phục CGI hoàn chỉnh.

36. NO: Sẽ mất một thời gian dài nữa các nhà làm phim mới sử dụng hình ảnh chụp cắt lớp của vải (Đoạn 5: bài viết nói việc sử dụng cho da mới mất nhiều năm, còn với vải/trang phục thì hiện đã có kết quả - "for the moment... look far more realistic").

37. A (light): Nhóm của Marschner đi đầu trong việc áp dụng hiểu biết về cách **ánh sáng** tương tác với da người.

38. F (mathematics): Mô hình CGI của họ dựa trên một ứng dụng mới của các nguyên lý **toán học**, vốn trước đây được dùng trong nghiên cứu khoa học khác (thiên văn học).

39. D (use): Tầm quan trọng của CGI đã dẫn đến việc **sử dụng** mô hình của Marschner bởi các xưởng phim kỹ xảo.

40. G (improvement): Mô hình của Marschner đã dẫn đến sự **cải thiện** của kỹ thuật điện ảnh.